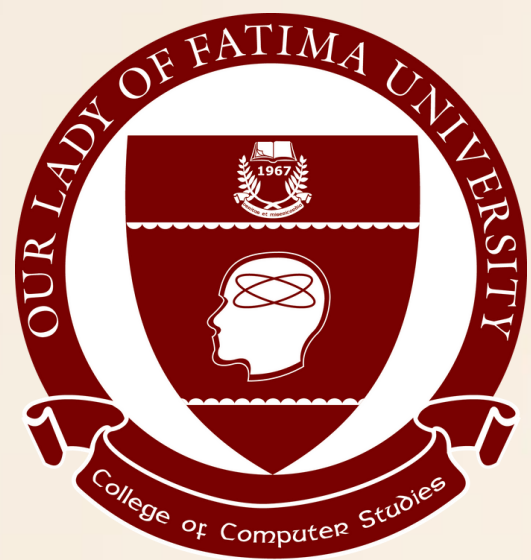




Module 3

Components of Computer System

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

INTRODUCTION



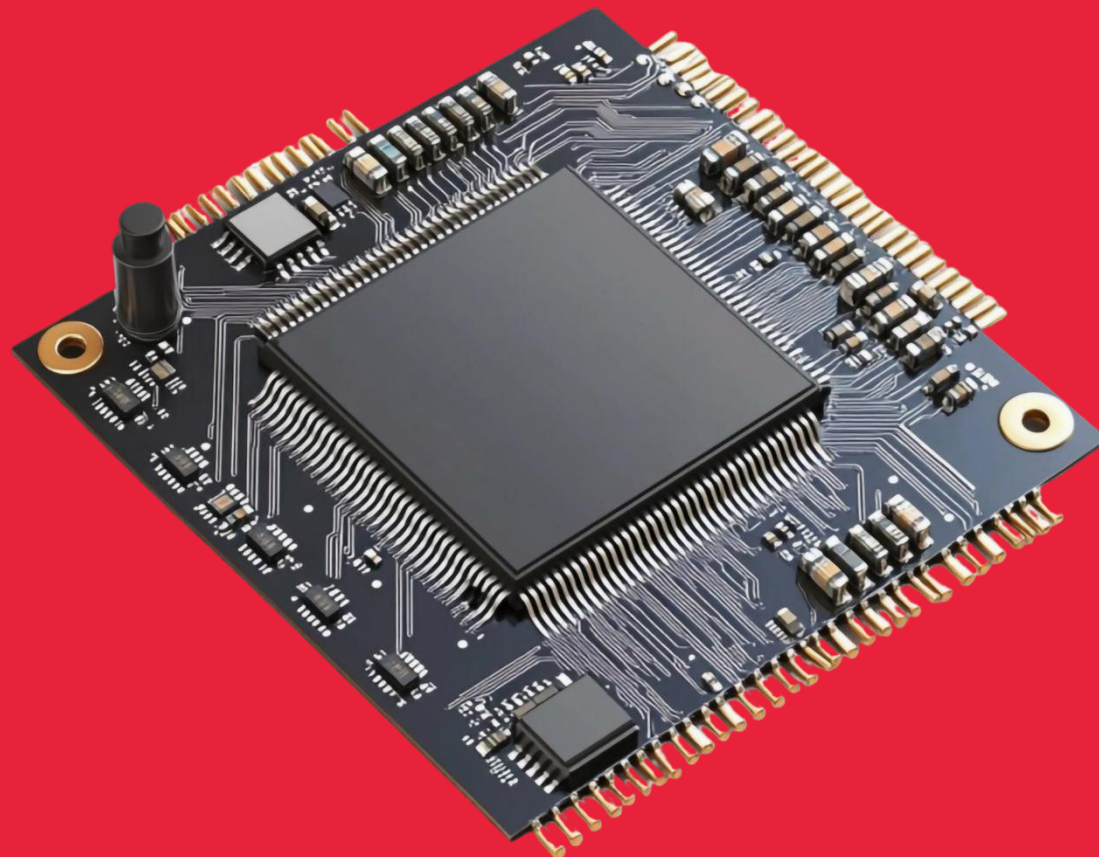
A computer system is made up of different components that work together to perform tasks efficiently. These components are classified into hardware, software, peopleware, and data. Each plays a vital role: hardware provides the physical structure, software gives instructions, peopleware refers to the users, and data is the information being processed. Understanding these components helps us see how computers function as powerful tools in everyday life.

MODULE 3

- 1. Basic Computer Architecture**
- 2. Input and Output Devices**
- 3. Storage Devices and Media**
- 4. System vs Application Software**



CENTRAL PROCESSING UNIT (CPU)



The CPU is often called the "brain of the computer" because it performs most of the processing operations. It interprets and executes instructions from programs and coordinates the activities of all hardware components.

Main Components of CPU

1. Control Unit (CU)

- Directs the flow of data within the computer.
- Interprets program instructions and tells other parts of the system what to do.
- Works like a “traffic controller” inside the CPU.

2. Arithmetic Logic Unit (ALU)

- Performs all arithmetic operations (addition, subtraction, multiplication, division).
- Handles logical operations (comparisons like $<$, $>$, $=$, AND, OR, NOT).

3. Memory Registers

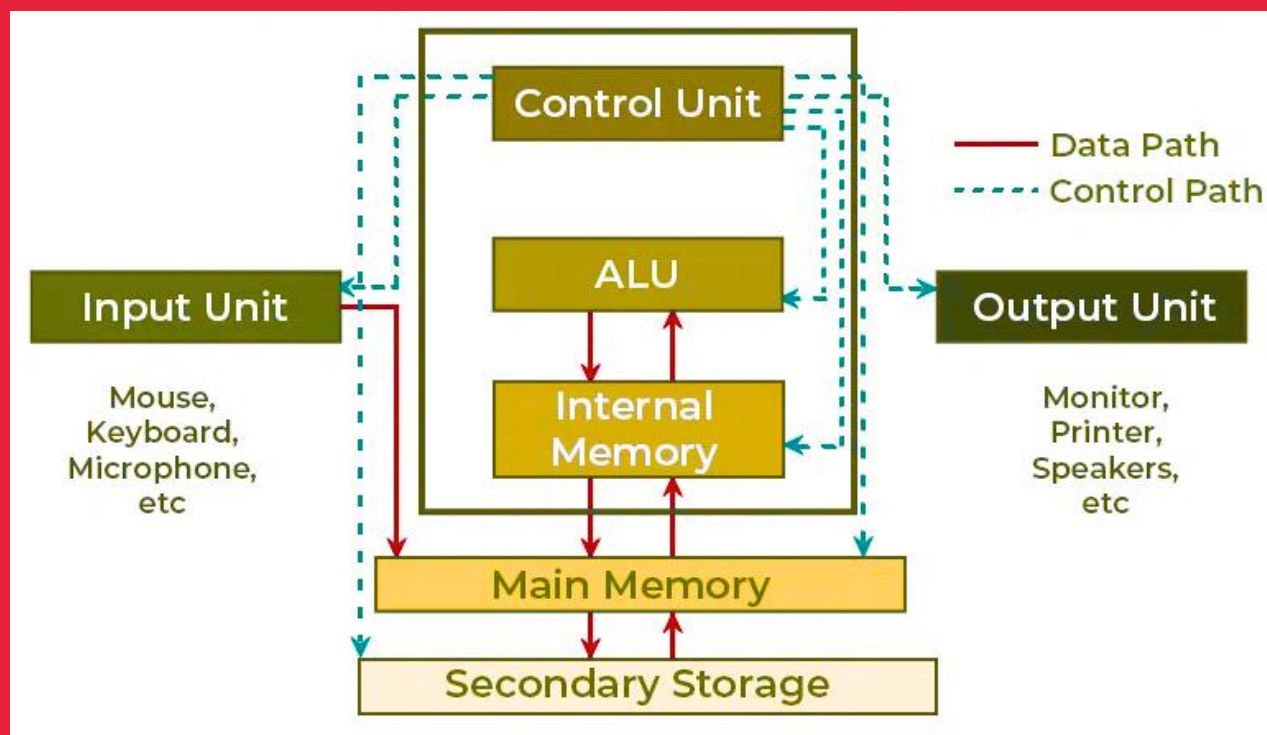
- Small, high-speed memory locations inside the CPU.
- Store temporary data such as instructions, addresses, and results of calculations.
- Examples: Accumulator, Instruction Register, Program Counter.

INPUT UNIT

The Input Unit is the part of the computer system responsible for taking data and instructions from the outside world (the user or another system) and converting them into a form that the computer can process (binary form).

Consists of input devices that are attached to the computer.

- Formed by attaching one or more input devices to a computer.
- A user inputs data and instructions through input devices such as a keyboard, mouse, etc.
- Used to provide data to the processor for further processing.



COMPONENTS OF EXTERNAL HARDWARE

INPUT HARDWARE – Text Input

These are devices that allow the user to enter textual data into the computer. The most common text input device. It uses a set of keys arranged in a standard layout (QWERTY).

INPUT HARDWARE – Terminals

A computer terminal is a hardware device (or software emulation) that allows a user to interact with a computer system.

Traditionally, a terminal was just a keyboard + screen connected to a mainframe or server.

TYPES of TERMINAL

1. Dumb Terminal

- No processing power (no CPU or storage).
- Only used for input (keyboard) and output (monitor).
- Relies 100% on the central computer/mainframe for processing and storage.

2. Smart Terminal

- Has limited processing power (can handle some functions like formatting, buffering, or minor computations).
- Can offload simple tasks from the main computer.

3. Intelligent Terminal

- Has its own CPU, memory, and storage.
- Can process data independently and sometimes operate without the main computer.
- Often capable of running applications locally.



COMPONENTS OF EXTERNAL HARDWARE



INPUT HARDWARE – Pointing Devices

A computer pointing device is a hardware input device that allows users to interact with a computer by moving a pointer or cursor on the screen.

1. Mouse

- A pointing device that detects two-dimensional movement relative to a surface.

2. Trackball

- A stationary pointing device with a ball that you rotate with your fingers.

3. Joystick

- A stick-like input device that moves in different directions.

4. Light Pen

- A pen-like device that detects light on the screen when touched.

5. Digitizing Tablet

- A flat surface with a stylus/pen used to draw or write digitally.

6. Pen-Based Systems

- Entire systems designed around pen input (e.g., stylus with touch-sensitive screens).

COMPONENTS OF EXTERNAL HARDWARE



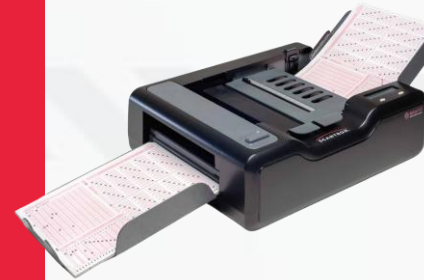
INPUT HARDWARE – Scanning Devices

Scanning devices are input devices used to capture images, text, or objects and convert them into a digital format that a computer can process, store, or display. Instead of manually typing or recreating data, scanners allow fast and accurate digitization of printed or physical content.



1. MICR (Magnetic Ink Character Recognition)

An input device that reads characters printed with special magnetic ink. Commonly used in banks for processing cheques.



2. OMR (Optical Mark Recognition)

A device that detects marks made on specially designed paper forms, such as shaded bubbles or checkboxes.



3. OCR (Optical Character Recognition)

A device/software that reads printed or handwritten characters and converts them into digital, editable text.



4. Fax Machine

A fax machine is a telecommunication device that allows documents (text or images) to be transmitted from one location to another through telephone networks by scanning and reproducing them on paper.

COMPONENTS OF EXTERNAL HARDWARE



INPUT HARDWARE – Scanning Devices

Scanning devices are input devices used to capture images, text, or objects and convert them into a digital format that a computer can process, store, or display. Instead of manually typing or recreating data, scanners allow fast and accurate digitization of printed or physical content.

1. Optical Cards

Plastic cards that store data on an optical surface, similar to how CDs/DVDs store information.

Example: Medical history cards for patients. National ID systems in some countries. Secure document storage (military, government records).

2. Smart Cards

Pocket-sized plastic cards with an embedded microchip (microprocessor or memory chip) that can store and process data.

Example: ATM/debit/credit cards. SIM cards in mobile phones. Access control cards (company IDs, hotel keys). E-passports and government IDs.

COMPONENTS OF EXTERNAL HARDWARE



INPUT HARDWARE – Voice Recognition System

(also called speech recognition system) is a technology that enables a computer, smartphone, or any device to receive, interpret, and understand spoken language. It converts human speech into text or commands that machines can process.

These systems are widely used in virtual assistants (e.g., Siri, Alexa, Google Assistant), customer service bots, dictation software, and security systems.

1. Audio-input device

- they record or play analog sounds

2. Video-input device

- signals that come from a VCR or a camera recorder

3. Electronic cameras

- they capture images in electronic form

4. Sensors

- they collect specific kinds of data directly from the environment

OUTPUT UNIT

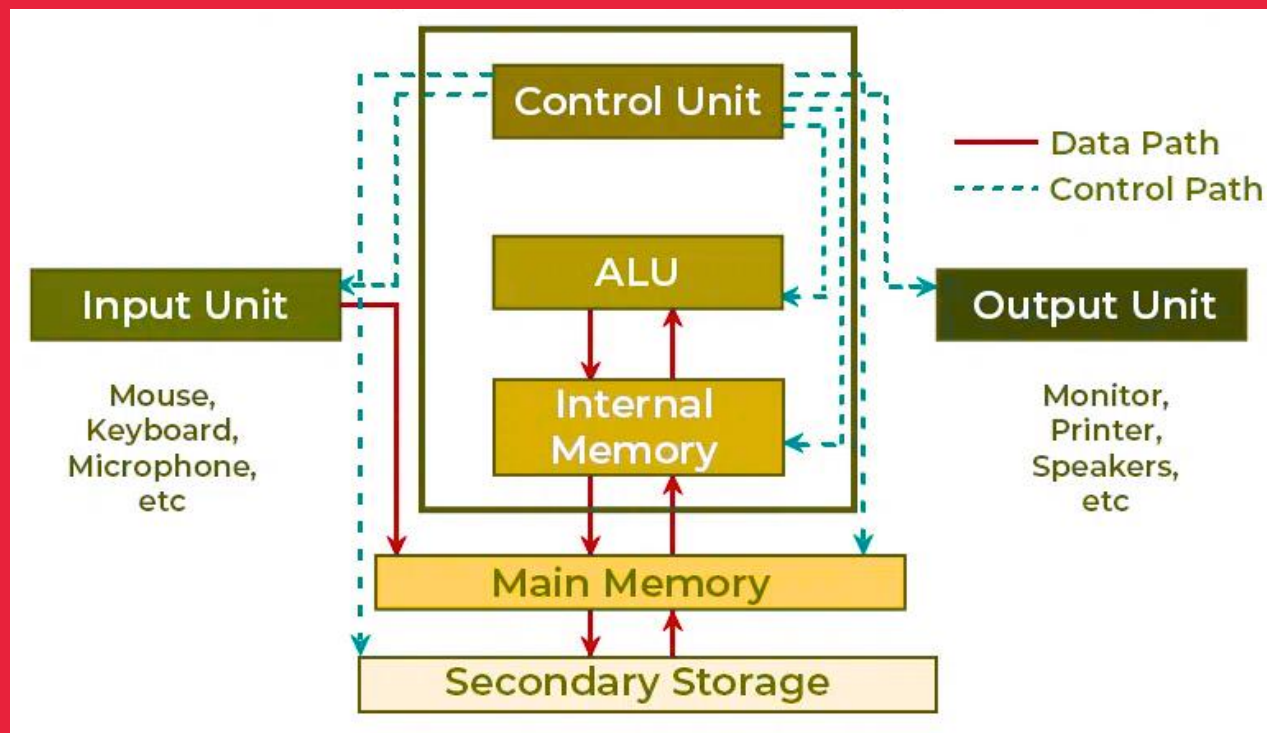
The Output Unit is the part of the computer that delivers the results of processing from the CPU and memory to the outside world, such as users, other computers, or external devices.

It converts the binary data coming from the CPU to human understandable form.

- output unit displays or prints the processed data in a user-friendly format.
- output unit is formed by attaching the output devices of a computer.
- output unit accepts the information from the CPU and displays it in a user-readable form.

Examples of Output Devices:

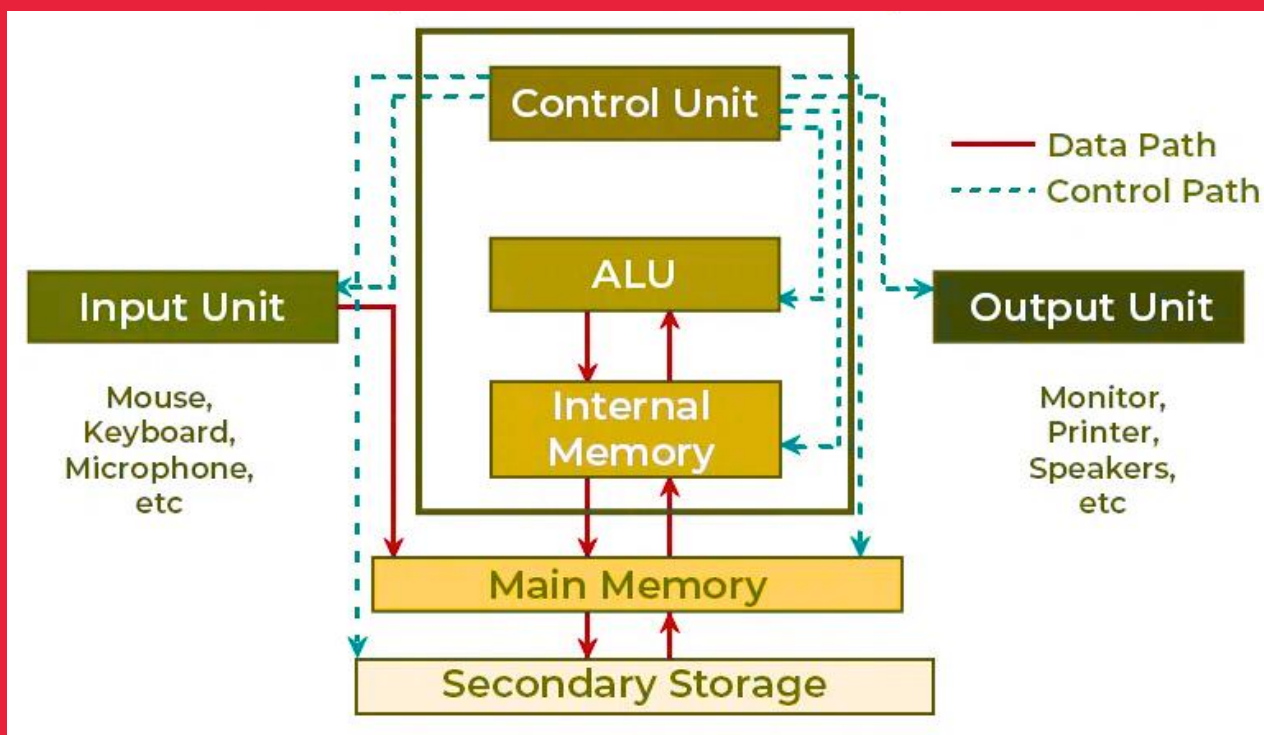
- Visual Output
 - Monitor / Screen (LED, LCD)
 - Projector
- Printed Output
 - Printer (Inkjet, Laser)
 - Plotter
- Audio Output
 - Speakers
 - Headphones
- Other Output
 - Actuators in robotics
 - Digital signals to other systems



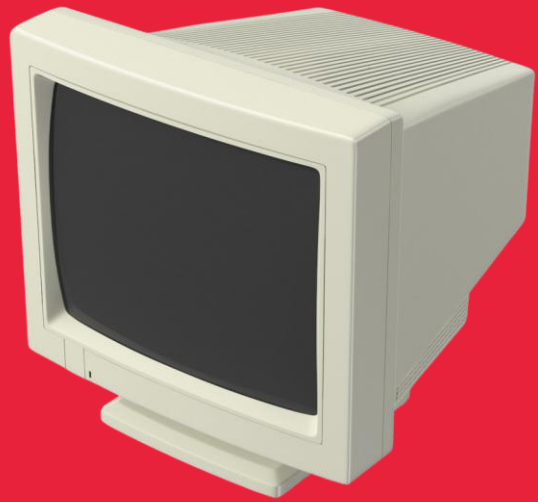
INPUT/OUTPUT UNIT

Some devices can perform both input and output functions.

- Touchscreen – accepts input through touch while also displaying output.
- External Drives (USB / HDD) – input (save data to computer) and output (retrieve data).
- Network Devices (Modem, Router) – send and receive data.



COMPONENTS OF EXTERNAL HARDWARE



OUTPUT HARDWARE – Visual Output Devices

The Output Unit is the part of the computer that delivers the results of processing from the CPU and memory to the outside world, such as users, other computers, or external devices.

1. CRT (Cathode Ray Tube) Monitor

An older type of display device that uses electron beams to excite phosphor dots on a glass screen to produce images.

Example: The traditional “box-type” computer monitors and old television sets.

2. Flat Panel Display (FPD)

A thin, lightweight display technology that replaced CRTs in most devices. It uses different technologies like LCD (Liquid Crystal Display), LED (Light Emitting Diode), and OLED (Organic LED).

Example: Modern LED TVs, laptop screens, tablet displays.

3. Projector

An output device that projects visual content from a computer or media player onto a large screen or surface.

Example: Multimedia projectors used in schools, conference rooms, or movie theaters.

COMPONENTS OF EXTERNAL HARDWARE



OUTPUT HARDWARE – Print Output Devices

A printer device is an output device of the computer that produces a hard copy (physical copy) of digital documents, images, or graphics on paper or other printable media.

1. Impact Printer

A type of printer that prints by striking an inked ribbon against paper. Similar to a typewriter — a mechanical head hits the ribbon, and the ink transfers to the paper. Mechanical, noisy, can make copies.

Example: Dot Matrix Printer, Daisy Wheel Printer, Line Printer

2. Non-Impact Printer

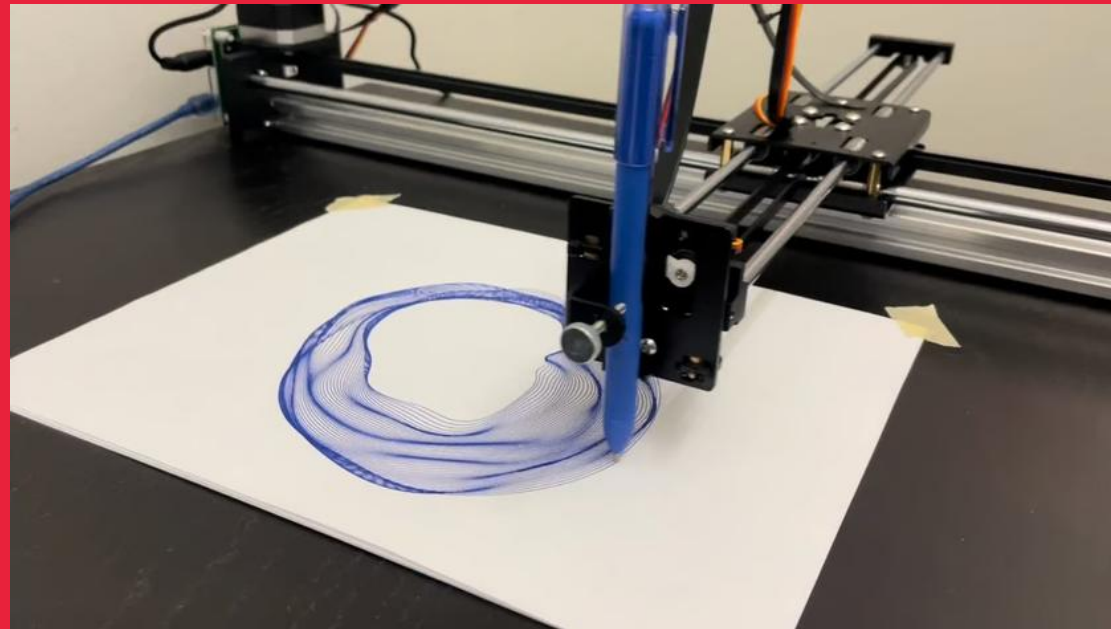
A type of printer that prints without physical contact between the print head and the paper. Quiet, high-quality, no physical striking.

Example: Laser Printer, Inkjet Printer, Thermal Printer

3. Plotter

A plotter is an output device used to produce large-scale, high-quality graphics, drawings, and designs on paper. Unlike regular printers, plotters can draw continuous lines (not just dots), making them ideal for CAD drawings, maps, blueprints, and engineering designs.

COMPONENTS OF EXTERNAL HARDWARE



OUTPUT HARDWARE – Print Output Devices

A printer device is an output device of the computer that produces a hard copy (physical copy) of digital documents, images, or graphics on paper or other printable media.

TYPES OF PLOTTER

1. Pen Plotter

Uses pens of different colors and thicknesses to draw directly on paper.

Example: Used in blueprints and technical drawings.

2. Electrostatic Plotter

Works like a photocopier — uses electrostatic charges on special paper, then applies toner (powdered ink) to create images. Good for printing large maps and charts.

3. Thermal Plotter

Uses heated elements to burn or melt images onto heat-sensitive paper. Commonly used for scientific data recording, ECG machines, and small technical drawings.

COMPONENTS OF EXTERNAL HARDWARE



OUTPUT HARDWARE – Audio Output Devices

Refers to computer hardware devices that produce sound, allowing users to hear audio from a computer or other digital device. These devices convert digital audio signals into audible sound waves. They are mainly used for listening to music, voice, alerts, sound effects, or any multimedia content.

1. **Speakers** – External or built-in devices that amplify and play sound for everyone to hear.
2. **Headphones** – Worn over or in the ears for private listening.
3. **Earphones / Earbuds** – Smaller, portable versions of headphones.
4. **Headsets** – Combination of headphones and a microphone, used for communication (e.g., gaming, online calls).

COMPONENTS OF EXTERNAL HARDWARE



OUTPUT HARDWARE – Multimedia Output Devices

Multimedia Output Hardware refers to the physical devices that present multimedia information (text, audio, images, video, and animation) from a computer to the user. These devices allow people to see, hear, and interact with multimedia content.

1. SMART tv

A modern television with internet connectivity and built-in apps (e.g., YouTube, Netflix).

Example: Home entertainment, online streaming, interactive presentations.

2. Multimedia Kiosk

A standalone interactive terminal with a touch screen, often found in malls, airports, or museums.

Example: Information booths, ticketing machines, self-service ordering systems.

3. Digital Signage Systems

Large electronic displays (LED/LCD screens) used to show dynamic content like ads, announcements, or schedules.

Example: Airports, train stations, shopping malls, schools, restaurants (digital menus).

STORAGE DEVICES AND MEDIA



Storage devices and media are components used to store, save, and retrieve data and information, either temporarily or permanently, in a computer or digital system.

Two Main Types of Memory

1. Primary Memory (Volatile)

Directly accessible by the CPU. Temporary storage; data is lost when power is turned off.

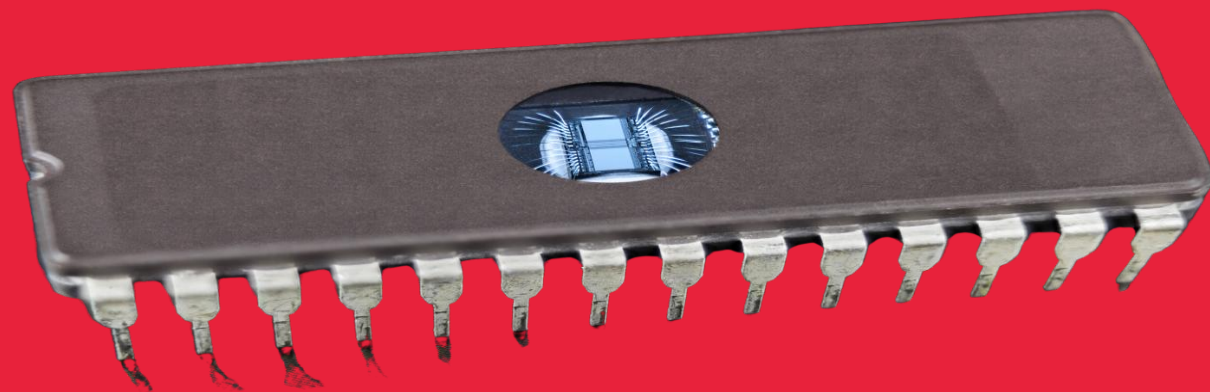
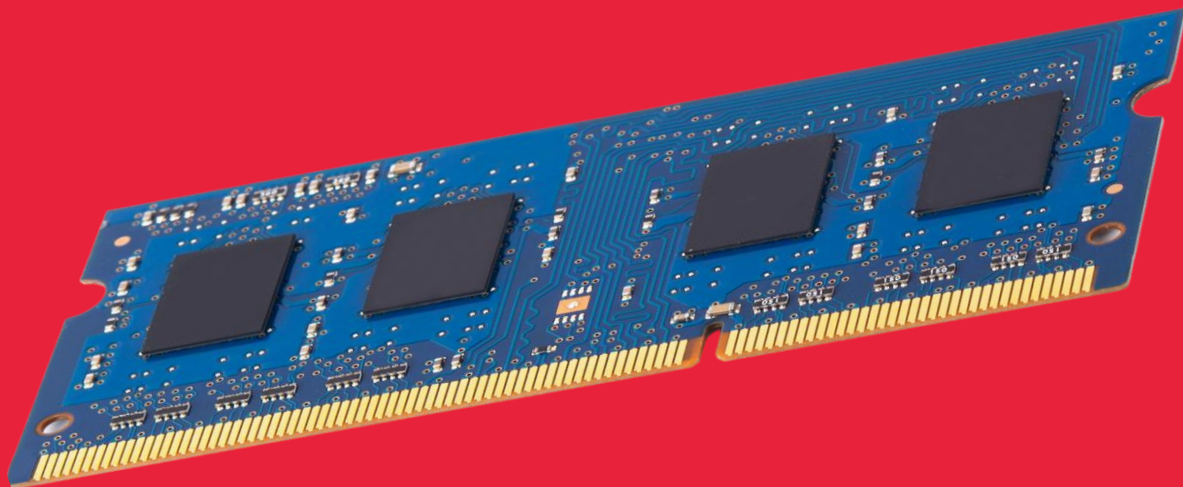
Example: RAM & Cache Memory

2. Secondary Memory (Non-volatile)

Permanent storage; data remains even when the power is off.

Example: HDD, SSD, Flash Drives, DVDs

STORAGE DEVICES AND MEDIA



Computer internal memory (also called primary memory or main memory) is the memory located inside the computer that stores data and instructions needed by the CPU while performing tasks.

Types of Internal Memory

1. RAM (Random Access Memory)

Volatile memory used for storing data and instructions while a computer is running. Allows both read and write operations.

2. ROM (Read-Only Memory)

Non-volatile memory that permanently stores essential instructions needed to boot the computer. Cannot be easily modified.

3. Cache Memory

Stores frequently used instructions and data for faster access.

4. Registers

Used to hold data temporarily during processing.

STORAGE DEVICES AND MEDIA



Computer External Memory (also called secondary storage or auxiliary memory) refers to memory devices and storage media that are outside the main computer memory (RAM) and are used to store data permanently or for long-term use.

Types of External Memory

1. HDD (Hard Disk Drive)

Uses spinning magnetic disks (platters) to read/write data with a moving arm (like a record player). Slower (because of moving parts). More fragile; can be damaged if dropped. Usually offers large storage at a cheaper price.

2. SSD (Solid State Drive)

Uses flash memory (like a USB drive) with no moving parts. Much faster (quicker boot-up, file transfer, and loading times). More durable; resistant to shock since there are no moving parts. Usually smaller than HDDs (120GB–4TB is common) and more expensive per GB.

REMOVABLE MEDIA DEVICES



Removable media refers to any type of storage device that can be easily inserted into and removed from a computer system without shutting it down. It is designed for portable data storage, transfer, and backup.

1. **Diskettes**
Portable magnetic storage (floppy disks).
2. **Optical Storage**
Uses laser technology to read/write (e.g., CD, DVD, Blu-ray).
3. **Magnetic Tapes**
Sequential storage using magnetic coating on tape.
4. **Paper Tape**
Early data storage medium (holes punched in paper).
5. **Continuous Forms**
Special paper (perforated, connected sheets) used in line printers.
6. **Computer Output Microfilm (COM)**
Converts computer output into microfilm/microfiche.
7. **USB Flash Drives**
Portable solid-state storage.
8. **Memory Card**
Solid-state storage, used in phones, cameras, etc.

CLOUD STORAGE



Cloud storage is a service that allows you to save data and files on remote servers instead of your computer's local hard drive or other physical storage devices. These servers are maintained by cloud providers (such as Google Drive, Dropbox, OneDrive, iCloud, or Amazon S3), and your data is accessible through the internet.

- **Google Drive** – Lets you store documents, photos, and videos online and access them from any device with your Google account.
- **Dropbox** – A popular cloud storage for file sharing and collaboration.
- **Microsoft OneDrive** – Integrated with Microsoft Office (Word, Excel, PowerPoint), useful for students and professionals.
- **Apple iCloud** – Used for backing up iPhones, iPads, and Mac data like photos, contacts, and files.
- **Amazon S3 (Simple Storage Service)** – A cloud storage solution often used by companies for large-scale data storage and backup.

SYSTEM SOFTWARE VS APPLICATION SOFTWARE

System Software

Software that manages and controls the computer hardware so other software and users can work with it.

Purpose: Acts as a bridge between the hardware and the user.

Examples:

- Operating Systems (Windows, macOS, Linux, Android, iOS)
- Utility Programs (antivirus, disk management tools, file compression tools)
- Device Drivers (software that makes hardware work, e.g., printer drivers)

Application Software

Software designed for users to perform specific tasks or solve particular problems.

Purpose: Directly helps the user accomplish work, study, entertainment, or daily tasks.

Examples:

- Microsoft Word, Excel, PowerPoint (productivity apps)
- Google Chrome, Firefox (browsers)
- Photoshop (graphics)
- Games, media players